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School-Based Social Skills Interventions for Youth With ADHD: A Systematic Review and Meta-Analysis

Journal of Attention Disorders
2026, Vol. 30(4) 427–448
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DOI: 10.1177/10870547251364578
journals.sagepub.com/home/jad
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Abstract

Objective: Children and adolescents frequently struggle with social skills performance in academic settings, which can negatively impact quality of life. As such, social skills training is frequently recommended; however, the effectiveness of school-based social skills training has not been synthesized. As such, the purpose of the current study was to evaluate the empirical evidence pertaining to both stand-alone and multimodal school-based social skills programs for students with ADHD, and to assess the effects of potential moderating variables. **Method:** A systematic review and meta-analysis was conducted of studies between the years of 1975 and 2023 following the recommendations of the Cochrane Foundation and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). **Results:** Seventeen studies, single-case, within group, and between-group designs met the established inclusion criteria, with 10 included in the meta-analysis. The findings revealed small to large effects; however, the overall meta-analytic effect size was negligible (ES = 0.09). The differences in social skills approaches, intensity, parent involvement, and setting did not moderate the effects.**Conclusion:** These findings underscore the necessity for ongoing exploration and refinement of social skills interventions tailored for youth with ADHD. (*J. of Att. Dis. 2026; 30(4) 427-448*)

Keywords
ADHD, social skills, training, programs

Introduction

ADHD is a neurodevelopmental disorder characterized by inattention, hyperactivity, and/or impulsivity (American Psychiatric Association [APA], 2013). Approximately 8.7% of children between the ages of 3 and 17 years old are diagnosed with ADHD (Bozinovic et al., 2021). Students with ADHD often experience difficulties in academic, behavioral, and social domains, which can negatively impact their overall quality of life (DuPaul & Langberg, 2015). Social skills deficits are a strong predictor of overall functioning, both in the short and long term, underscoring the importance of addressing these challenges (Al-Yagon et al., 2020; Bagwell et al., 2001; Fabiano et al., 2018; Rutherford et al., 2008).

Behaviors demonstrated by students with ADHD are frequently incompatible with social competence, especially in academic settings. As such, they often experience social skills deficits (i.e., behaviors and abilities necessary for successful interactions with others; Hoza, 2007). Nonetheless, evidence has suggested that this is not always due to knowledge of social skills, but instead performance (Aduen et al., 2018). Specifically, individuals with ADHD tend to interrupt, not follow directions, be forgetful in daily activities, and demonstrate behavioral dysregulation when strong emotions are present (Zoromski et al., 2015). As such, individuals with ADHD are more likely than their peers without ADHD to be socially rejected (Strine et al., 2006). Consider the student with ADHD assigned to a group project. The student may struggle to stay on task, talking excessively, and interrupting others (Hoza, 2007; Wehmeier et al., 2010). The student is likely to become distracted, distract the group, and less likely to complete the task on time. These behaviors interfere with interpersonal relationships and academic progress. In this study, we refer to social skills deficits as both acquisition and performance of social skills.

In addition to the immediate effects of social skills deficits, the associated peer dislike and rejection can lead to

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Correction (September 2025): The article has been updated with the minor data correction in the Table 1.

symptoms of anxiety (Mulraney et al., 2018; Sciberras et al., 2014), depression (Eadeh et al., 2017; Feldman et al., 2017; Mrug et al., 2012), and substance use and abuse (Greene et al., 1997). For example, a correlation has been found between parent-reported (Karustis et al., 2000; Mikami et al., 2011; Mulraney et al., 2018; Sciberras et al., 2014), and child-and teacher reported (Karustis et al., 2000; Mikami et al. 2011) anxiety and peer rejection. These studies provide evidence that adolescents who have ADHD and struggle with social skills are more likely to be diagnosed with comorbid conditions that require additional support.

Although 16.5% of youth have at least one mental health disorder that is likely to interfere with social functioning, only 49.4% receive services (Whitney & Peterson, 2019). This is especially true for youth from underserved communities, including those of color and low-income families (Slobodin & Masalha, 2020). One means of increasing supports and services is to provide them in school (Kappi et al., 2022). However, researchers and practitioners have noted that the low ratio of counselors to students in schools might interfere with implementation (Donohue et al., 2022). Fortunately, social skills training (SST) can be implemented by teachers in classrooms along with school counselors (Bauminger, 2002). Furthermore, collaborating across professions (e.g., different teachers, counselors, occupational therapists) increases the potential to generalize skills taught in a classroom or small group setting (Powell & Sable, 2001).

School-Based Social Skills Training Programs and Interventions

SST programs are designed as stand-alone interventions that focus on specific skill-building or multimodal interventions that integrate interventions across areas of impairment (e.g., academic, social, and behavioral; Evans et al., 2011; Willis et al., 2019). However, the differential effectiveness is not known.

Although not focused on school-based services in the last 5 years, three systematic reviews and meta-analyses have synthesized the literature on the effects of training aimed at improving the social skills of children and adolescents with ADHD. Overall, SST resulted in inconsistent findings across study design and outcome measures, with observed effects ranging from null to moderate. Randomized controlled trials (RCT) tended to result in no to low effects, whereas within-subject designs demonstrated small to moderate effects. Furthermore, parent ratings were more likely to demonstrate positive effects than teacher ratings. Willis et al. (2019) summarized the findings of a systematic review with 16 group design studies as individual study authors "reported at least initial evidence to support the efficacy of stand-alone SST aimed at reducing social impairment" (p. 355) and noted inconsistent findings across studies with variations in reports from parents, teachers, and peers. Morris et al. (2021) calculated an overall effect across four RCTs and four between-group studies. They reported that the RCTs found no differences in social functioning between treatment and control groups as reported by parents (ES = −0.08). Nonrandomized studies demonstrated improvements in social functioning from baseline to postintervention by parent report with a moderate effect (ES = 0.57) and no effect by teacher report (ES = 0.04). Powell et al. (2022) calculated the overall effect with eight RCTs (with comparison groups of treatment as usual only) and found a small effect for teacher (ES = 0.32) and parent (ES = 0.39) ratings.

Thirty-two unique studies were included across all reviews, with only 10 conducted in schools or analogous settings (e.g., Summer Treatment Programs, After School Programs). Willis et al. (2019) focused on stand-alone SST in clinical settings, and Powell et al. (2022) focused on stand-alone and multimodal interventions with a focus on the psychoeducational components in clinical settings (six of eight studies). Morris et al. (2021) focused on nonpharmacological multimodal interventions in various settings, including schools. Furthermore, none of the three reviews included single case design studies, a typical design used in school-based research (Parker et al., 2006). Additionally, although differences were noted between RCT and within-group studies and parent and teacher ratings, none of the prior studies evaluated moderator effects. As such, our goal was to review the school-based intervention studies, synthesize the literature, and calculate the effectiveness of stand-alone and multimodal SST programs implemented in schools, and evaluate potential moderators.

Reviews and meta-analyses of clinic-based SST interventions have demonstrated positive outcomes; however, they also highlight a gap in research synthesis regarding the effectiveness of SST implemented in school settings. The current literature does not provide researchers or practitioners with information about the effects of SST implemented in schools.

Examining school-based SST is essential for several reasons. First, the generalization of skills from clinical environments to school contexts is often limited, as the structure and routines of clinics are different than school settings. In contrast, when SST is implemented in schools, instruction and practice occur in authentic settings, such as classrooms, recess, and cafeterias, where students are more likely to use the skills and reinforcement from peers and adults is more likely to occur. This setting also enables educators to provide immediate feedback, which is likely to enhance skill acquisition and performance (Ajogbeje, 2023). As such, intervention effects may be different in school settings compared to clinics.

Additionally, delivering SST in schools helps address systemic barriers to access, including transportation difficulties, limited availability of services, and high clinical care costs (Eiraldi et al., 2015). By embedding interventions in the school day, educators can reach students who might otherwise be unable to receive services, thereby promoting educational equity. Finally, the natural interplay between learning and behavior in school settings reinforces the relevance of SST. When social skills instruction is integrated into daily academic and social routines, it is likely to support classroom engagement, peer and adults relationships, behavioral regulation, and academic success.

Potential Moderators

Although not analyzed in prior meta-analyses, potential moderators emerged from findings and discussion within the literature. Several prior meta-analyses (e.g., Morris et al., 2021; Powell et al., 2022; Ros & Graziano, 2018) mentioned challenges with assessment. First, differences were found between parent and teacher ratings of social skills. Powell et al. (2022) noted that parent ratings were more likely to demonstrate a significant positive effect than teacher ratings. It is noteworthy that each intervention reviewed by Powell and colleagues included some form of parent training, suggesting that parents were invested in the outcomes and that blinding raters to the intervention was not possible. Morris et al. (2021) noted that the results of RCTs were similar regardless of rater. However, the results of non-randomized studies indicated that parents rated improved social functioning while teachers and students did not. As such, in conjunction with the exploration of SST outcomes, it is essential that we explore the impact of the reporters of skill when rating scales are the prime measure.

Second, some have questioned (e.g., Ros & Graziano, 2018) potential differences between findings when dependent measures were distal or broadband scales (e.g., BASC, SSIS). Specifically, questions regarding outcomes evaluated by tools that measure distal multiple dimensions (Raiker et al., 2017) versus proximal or narrowband, which measure just one proximal dimension (e.g., ADHD-specific rating scales; J. Pelham et al., 2005) have been questioned. For example, if a social skills program is used to teach problem-solving (e.g., Webster-Stratton & Reid, 2008), then direct observation or a single or multiple-item scale directly related to the use of problem-solving skills (i.e., narrowband) might be more sensitive to change in behavior associated with the intervention than a broadband scale.

The third was the type of control group (i.e., passive, active). The purpose of a control group is to account for influences on outcomes that are not directly associated with the intervention being tested. If the control group is active, some researchers argue that the potential for placebo effects is minimized, as the key is participant expectation (i.e., to improve based on the intervention). As such, comparing the effects of an intervention to the effects of a passive or no-treatment control group could potentially introduce confounding effects caused by differences in participant expectations (Boot et al., 2013), suggesting that effect sizes from studies with passive controls would result in larger effect sizes. For example, Noble et al. (2019) found that the effect of shared reading on language development was moderated by type of control group with negligible effect for studies with active control groups and larger for studies with passive controls. As such, using an active control group that receives support, or intervention will likely lead to lower effect size estimates than using a business-as-usual group that continues to receive services provided in the context without intervention.

The fourth was the type of intervention (i.e., stand-alone, multimodal). Stand-alone interventions target one outcome (e.g., social competence) through one intervention (e.g., social skills training groups). Alternatively, multimodal interventions target multiple outcomes (e.g., academic, behavioral, social) across numerous domains through various interventions (e.g., homework assistance, token economy, social skills training groups). Morris et al. (2021) mentioned that among the various interventions included in their meta-analysis, only one intervention focused only on the social domain, with others also targeting vocational, organizational, or others. Researchers suggest that multimodal interventions designed to target different areas of impairment (e.g., organization, academic functioning) have the potential to secondarily impact social skills due to the influence on protective factors, such as peer acceptance (Mrug et al., 2007). This introduces variability in the calculation of overall effects that quantify the extent to which the observed improvements in social functioning can be attributed to the intervention. As such, the type of intervention may moderate the overall effects.

Finally, the age and/or grade level of the students in the study have the potential to impact effectiveness. It is possible that age impacts the outcomes of SST, as maturity might affect how students engage with and benefit from social skills instruction. Social functioning becomes increasingly important as children move into adolescence, as peer relationships gain significance (Gresham et al., 2001). Furthermore, SST programs often require mature cognitive functioning, such as the ability to reflect on one's behavior, infer others' emotions or intentions, and apply learned strategies across settings (Merrell & Gimpel, 2014). As such, results of studies with younger children may result in smaller effects than those with older children or adolescents.

Need for Current Review and Meta-Analysis

As such, the purpose of the current study was to systematically evaluate the empirical evidence for school-based stand-alone and multimodal interventions that include SST for students with ADHD and evaluate potential moderators. Specifically, this systematic literature review and meta-analysis addressed the following research questions:

RQ1: What was the overall effect of social skills training implemented in schools on outcomes for students with ADHD? What was the heterogeneity in effects within and across studies?
RQ2: What was the scientific quality of studies of social skills training implemented in schools?
RQ3: To what extent did study characteristics (i.e., rater, measures, type of control group, type of intervention, age/grade level of participants) moderate the effect of social skills training?

Method

This systematic review and meta-analysis were conducted following the recommended procedures of the Cochrane Foundation (Henderson et al., 2010) and two methodological guidance papers (Alexander, 2020; Pigott & Polanin, 2020). Additionally, we consulted the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Standards (Page et al., 2021) to provide transparent and clear reporting of our methodology and results. The study was conducted in three phases: (a) systematic search and study selection, (b) data coding, and (c) data analysis. The following sections are organized accordingly.

Phase 1: Study Selection and Systematic Search

In the following sections, we outline the criteria for including source studies in this meta-analysis and describe our search procedures.

Inclusion Criteria. Studies were included that were (a) conducted between 1975 and 2023; (b) published in English; (c) published in peer reviewed journals; (d) used group or single case design; (e) included participants with a diagnosis of ADHD, at risk for ADHD, or with ADHD and comorbid condition; (f) included participants and setting in primary and secondary schools (K-12), or analogous settings (e.g., a summer treatment program, after school program) designed to align with typical classrooms; (g) examined the effects of a social skills intervention (i.e., designed to increase social competence, prosocial skills, interpersonal skills, and/or decrease social impairment, disruptive, or offensive behaviors in social settings; Gresham et al., 2004) as indicated by the authors, (h) included an outcome measure of social skills /social competence with scores disaggregated from other outcomes; and (i) was not a study of the effects of medication alone or as a comparison group. We included studies that were only implemented in schools or analogous settings. As such, studies that included both clinic and school components were excluded from inclusion.

We defined outcome measures of social skills as those designated to measure social skills (e.g., Social Skills Improvement Rating Scales; Gresham & Elliot, 2008) per the authors, with scores of social impairment included and disaggregated by the author (e.g., Impairment Rating Scale [IRS] Fabiano et al., 2006) or determined to be a social skill measure by the authors of the current paper. Specifically, when outcomes were not clearly defined as social outcomes, two authors reviewed the definition provided by source study authors and determined whether we agreed that the measure was an assessment of social skills. For example, Sibley et al. (2011) measured "inappropriate behavior during business meetings." As a team, we agreed that in this study inappropriate behavior was a social behavior due to the author definition, as "any verbal or nonverbal social interaction that would typically annoy others in the group or decrease the popularity of the adolescent." (p. 582). Villodas et al. (2013) included social development grades as an outcome. The authors reported that "the participating school district utilizes a standards-based report card system that requires teachers to rate students on grade-specific expectations for social development, such as "Respects self and others," "Resolves conflicts," and "Follows directions." In another example, Evans et al. (2004) disaggregated scores on the CIS that we agreed met the definition of social skills (i.e., relationships with playmates, sibling, parents, family).

For the meta-analysis, we further restricted the inclusion criteria. We restricted the study type parameter to only include group design studies with a minimum of two groups comparing different conditions. We excluded pre-post designs with only one group due to limitations in inferring causal relations and the inability to report a standardized effect size that was comparable to the included between-group studies. We excluded single-case designs because we identified only two studies, and one of these studies did not meet the requirements for using an analogous effect size metric (i.e., between-case standardized mean difference; see Hedges et al., 2013) in between-group design studies.

Search Procedures. Following procedures of the Cochrane Foundation (Henderson et al., 2010) and referring to methodological guidance (Alexander, 2020), an electronic database search was conducted. Additionally, we repeated the search, including the years 2022 and 2023. We initiated our search process by selecting databases that would be likely to include journals that publish articles fitting our inclusion

criteria: (1) Education and Resource Information Center (ERIC), (2) Academic Search Premier, and (3) PsycINFO. We searched each database using the following Boolean string: ("*social skills training*" OR "*social skills instruction*" OR "*social skills intervention*" OR "*peer problems*" OR "*social relationships*" OR "*friendships*" OR "*social competence*" OR "*social cognition*" OR "*problem solving*" OR "*problem-solving*" OR "*multimodal*") AND ("*school*" OR "*school-based*") AND ("*attention deficit hyperactivity disorder*" OR "*ADHD*"). After our initial search, we conducted an ancestral review of the citations of the included studies and added two studies recommended by reviewers. Our search was conducted in June 2023.

Phase 2: Coding

Each study was coded based on the following categories: (1) study characteristics, (2) program or intervention characteristics, (3) Council for Exceptional Children (CEC, 2014) quality indicators, and (4) study outcomes, measures, and means of result reporting.

Study characteristics included: (a) author; (b) year; (c) diagnosis; (d) setting; (e) name of intervention; (d) number of participants, (e) grade level; (f) age; (g) gender, and (f) implementer. Program and intervention characteristics included the content addressed in the interventions (e.g., peer relationships, communication) and the specific components implemented (e.g., parent training, role-play).

The *Council for Exceptional Children Standards for Evidence-Based Practices in Special Education* (CEC, 2014) were used to evaluate the study quality of both group and single-case designs. For group design studies, 24 characteristics categorized across eight indicators were coded. Each indicator was coded dichotomously with response options indicating whether the criterion was met or not. For single-case design studies, 22 characteristics categorized across eight indicators were coded. Each indicator was coded dichotomously with response options indicating whether the criterion was met or not. If a study met all eight indicators it was coded as meeting standards, and if a study met fewer than eight indicators it was coded as not meeting standards.

Study outcomes, for both group designs and single-case designs, were coded as academic, social, behavior, symptomology (ADHD, other), and/or executive functioning (e.g., self-management, organization, planning). Measurement tools were coded directly from study author reports (e.g., observation, SSIS, IRS) in a textual format. Regarding statistical reporting for group designs, we extracted means, standard deviations, and sample sizes for each group, across all measures, at pretest and posttest. If means and standard deviations were not provided, other information was extracted (e.g., F-statistic, *p*-value) to estimate an effect size index. For single-case design studies, a screenshot of the time-series graph was uploaded to a graph digitizer with demonstrated evidence of reliable and valid estimates of raw data values (Rohtagi, 2022). The *x*-axis and *y*-axis value were estimated for each data point and coded for condition (e.g., baseline, intervention, maintenance).

Training and Inter-Rater Agreement on Coding Activities. Prior to independent coding, the second author (doctoral level with extensive experience) trained the members on all coding manuals. Specifically, the author reviewed the codes with coauthors, explained the codes in detail, modeled coding procedures, and conducted exercises in which raters identified one example and non-example of each coded variable. If codes were unclear, the team deliberated until a consensus was reached.

The members of the team who coded articles were three Ph.D. level researchers, one Ph.D. candidate, and three doctoral students. During training the team reviewed five studies together to define and clarify codes as needed. Initial training was conducted on four separate occasions (one training per criterion) for an hour each. This was followed by four practice sessions. Prior to the practice sessions, each member of the research team, independently coded five articles. Each article was independently coded by three raters simultaneously for discussion and clarification of codes when questions arose. During the meetings, the entire team reviewed and discussed disagreements. No disagreements were found for study characteristics and outcomes. As such, simple agreement was 100%. Disagreements were found within the CEC criteria; specifically, when the term "sufficient" was used in the description of the indicators (e.g., sufficient description of . . .). When disagreement occurred, raters discussed until 100% consensus was reached. If consensus was not reached, the second author made the final decision. Finally, two study authors independently coded 62% of all studies. Simple agreement was 96%.

Phase 3: Data Analysis

The data analysis process involved four phases. The first phase was a descriptive evaluation of included studies ($k=17$). The second phase was restricted to the single-case design studies ($k=2$). We conducted visual analysis and reported effect size indices for these studies (see below). The third phase was restricted to one group, pre-posttest design studies ($k=9$). We computed standardized mean difference effect sizes comparing changes in scores from pre-post using Cohen's *d*. The fourth phase was the meta-analytic model (see below), which was restricted to the studies meeting inclusion criteria defined above ($k=9$).

Descriptive Evaluation. First, we reported the coded variables for each study. Next, to evaluate the sample of studies included in our review across variables, we reported

descriptive data (i.e., frequency, percent) at the variable level for all coded variables. Then, we coded the scientific quality of each study using the *Council for Exceptional Children Standards for Evidence-Based Practice in Special Education* (CEC, 2014).

Visual Analysis and Quantitative Analysis of Single-Case Design Studies. For the single-case design studies ($k=2$), we conducted visual analysis to evaluate the existence of a functional relationship between the independent and dependent variable(s). Our visual analysis process was informed by the What Works Clearinghouse (WWC) protocol (Kratochwill et al., 2010) with guidance from Lane and Gast (2014). Specifically, we analyzed data by visually inspecting the: (a) level, variability, and trend of baseline data; (b) level, variability, and trend of intervention data; (c) and changes in level or trend between baseline and intervention, number of overlapping data points between baseline and intervention, immediacy of effect, and consistency across similar phases. We identified the number of basic effects observed (i.e., individual AB) and calculated the ratio of positive to negative or null effects.

For the single-case design studies, we reported a nonoverlap metric (i.e., Tau-U; Parker et al., 2011) and the between-case standardized mean difference for the one study that fit criteria (Hedges et al., 2013).

Quantitative Analysis of One Group Pre-Post Designs. For the one group pre-post test designs we reported Cohen's *d* as presented by the source study authors or calculated it using the reported means and standard deviations,

Meta-Analysis. We included quasi-experimental and RCTs, if they included a minimum of two groups of students assigned to different conditions ($k=9$, effect sizes = 81). The standardized mean difference effect size was used as the effect size metric (see Lipsey & Wilson, 2001). This involved the mean difference between the posttest scores from the target experimental condition versus the target comparison condition divided by the pooled standard deviation of the treatment and comparison groups' posttest scores. We corrected for small sample sizes by converting Cohen's *d* to Hedges' *g*. In cases where the means and standard deviations were not provided by authors, we estimated the effect size estimate using the information provided (see Wilson, n.d.).

Multiple studies used more than one social outcome measure, resulting in multiple effect size estimates per study. Similarly, some studies included multiple subgroups (e.g., boys in treatment A versus boys in treatment B; girls in treatment A versus girls in treatment B), which led to multiple effect size estimates per study. An issue arises if each effect size is treated as independent; effect sizes from the same study are likely to be more correlated than they are to effect sizes from other studies. If ignored, this dependency can lead to bias estimates in treatment effects. Thus, we provide two models for readers to interpret. In our first model, we built a random effects meta-analyses model that used the inverse variance weighted mean at the study level. However, this model loses valuable pieces of information, specifically, the heterogeneity of effect sizes at the study level. Thus, the final model built for interpretation was a multilevel, multivariate model with robust variance estimate (see Hedges et al., 2013).

Although the average effect may be of interest, a key strength of meta-analysis is to investigate how effects vary across studies and to investigate the nature of this variation, or heterogeneity. To investigate this heterogeneity, we started by creating a forest plot of effect sizes from each included study. Next, we computed the standard deviation of the true effects (τ) and the variance in true effects (τ^2). Third, we computed a prediction interval, which is the range of the true effects if a new study were added in the future (IntHout et al., 2016). Fourth, we used empirical Bayes estimates to generate a distribution of effect sizes that uses the estimates generated from the studies included in the meta-analysis and the average effect size across studies. Last, we conducted moderator analyses on five specific variables of focus: (a) age/grade of participants, (b) active versus business-as-usual comparison group, (c) the role of the person providing ratings for outcome measure, (d) intervention type (i.e., multimodal vs. stand-alone), and (e) broad versus narrow band outcome measures.

Publication Bias. Publication bias occurs when the published literature is systematically similar across specific variables from the entire population of literature on the phenomenon of interest. The most general case discussed related to this phenomenon is that published literature is more likely to include interventions demonstrating statistically significant effects compared to the entire population of literature on the topic. In systematic reviews and meta-analysis this can lead to overestimates of intervention effectiveness. To investigate publication bias, we started by generating two funnel plots, which plot the standard error (*y*-axis) versus the effect size estimate (*x*-axis). The first funnel plot included all 81 effect size estimates (see Supplemental Figure 1), and the second funnel (see Supplemental Figure 2) plot included the weighted mean effect of the nine included studies. We then used Egger's regression test as a quantitative approach to supplement our visual analysis. To further explore the potential nature of publication bias we analyzed the distribution of effect sizes across different *p*-value buckets ($<.025$; 0.025 to 0.50; and 0.50–1.0) along with the estimated effect size estimate at each bucket.

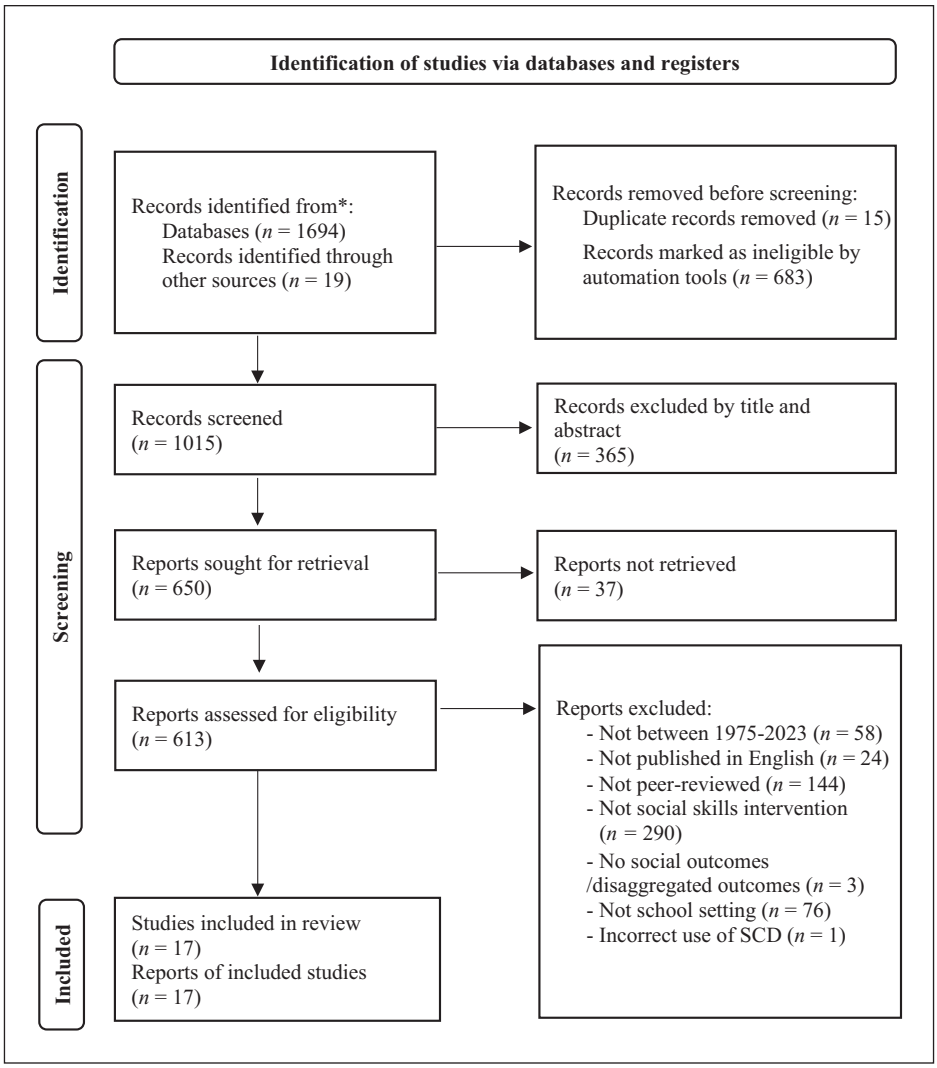


Figure 1. PRISMA Flow Diagram.

Results

The results are organized in four sections: (a) study identification and characteristic; (b) descriptive results of included studies; (c) effects at study level; and (d) meta-analysis results.

Study Identification and Characteristics

Results of the search can be found in the PRISMA Flow Diagram (see Figure 1). A total of 1,713 records were identified for the systematic review, including 1,694 from database searches and 19 from other sources. After removing 15 duplicate records and 683 records flagged as ineligible by automation tools, 1,015 records remained for title and abstract screening. Of these, 365 records were excluded, and 650 reports were sought for full-text retrieval. Thirty-seven reports could not be retrieved, leaving 613 reports assessed for eligibility. A total of 596 reports were excluded for the following reasons: not published between 1975 and 2023 ($k=58$), not in English ($k=24$), not peer-reviewed ($k=144$), not focused on social skills interventions (e.g., medication-only, assessment, or review articles; $k=290$), no disaggregated or relevant social outcome measures ($k=3$), not conducted in a school setting ($k=44$), incorrect use of single-case design ($k=1$), and other analytic or methodological limitations ($k=32$). Seventeen studies met all inclusion criteria and were included in the final review.

The studies included a total of 2,195 participants with 1,288 in treatment groups and 907 in nine control groups. Of the 2,195 participants, 1,452 (66.2%) were male and 743 (33.8%) were females. Some studies reported race alone and others reported ethnicity and race, as such the following

numbers do not sum to the total participants across studies: 1319 (60.1%) White; 247 (11.3%) Black; 192 (8.7%) Asian, 7 (0.3%) Native American, 79 (3.6%) Other; and 414 (18.9%) Latinx/Hispanic.

Of these 17 studies, 2 used a single-case design, 6 used a within-subject design (one group pretest-posttest design), and 9 used a group design with a minimum of two comparative groups. The nine group design studies included 81 effect sizes and were included in the meta-analysis.

Fourteen instruments were used to measure social outcomes. Within those instruments, outcomes were measured as: (a) social problems; (b) social skills; (c) social functioning (e.g., relations with peers, social impairment); (d) social behavior (negative, positive, inappropriate, appropriate); (e) relationships or interpersonal behavior (i.e., with children, peers, parents, teachers, siblings, playmates); (f) peer nominations/ratings (i.e., positive/negative nominations, reciprocated friendships, peer sociometric ratings; peer interactions; message from peers); (g) specific social skills (i.e., communication, cooperation, assertion, responsibility, empathy, engagement, self-control); (g) isolated play; (h) social development grades; (i) student report of parent support; and (j) student report of teacher support.

Studies were conducted in schools ($k=9$) and analogous settings ($k=7$; i.e., summer treatment programs [STP], after-school programs), and both ($k=1$; i.e., school and analogous settings; see Table 1). Several studies were conducted with students across schools, including elementary ($k=7$), elementary/middle school ($k=1$), middle school ($k=6$), middle/high ($k=1$), and high school ($k=2$).

Scientific Quality. Of the 17 studies, 10 met all eight CEC quality criteria (see Table 2). Four met seven out of eight criteria, one met six criteria, and two met five out of eight criteria. Four studies (Barney et al., 2022; Evans et al., 2004, 2005; Schultz et al., 2017) did not report fidelity (i.e., adherence, integrity) to procedures. Three studies (Barney et al., 2022; Evans et al., 2005, 2011) did not document attrition and one single-case design study did not document three opportunities to demonstrate an effect at three different points in time (Sibley et al., 2011). Two studies (i.e.., Barney et al., 2022; Evans et al., 2005) did not provide sufficient details of interventionists' training for replication. One study (Evans et al., 2004) did not report reliability of the measurement.

Interventions. Three stand-alone interventions and variations of three multimodal interventions were evaluated in the 17 studies. In the following sections, we describe the interventions, including the content, components, and summary of evidence within the categories of stand-alone and multimodal interventions.

Stand-Alone. Stand-alone social skills training programs are defined as training programs aimed only at improving and measuring social skills.

Social Skills Training Program. Colton and Sheridan (1998) investigated the effectiveness of a social skills training program. The intervention involved child training and conjoint behavioral consultation (CBC) with parents and teachers to support the child in demonstrating cooperative peer interactions. Procedures involved: (a) a problem identification interview with the child's mother and teacher; (b) problem analysis conversation between the consultant and the teacher and parent, identify target behaviors, and planning skills to be taught in the program. Child skills training involved: (a) coaching and role-play; (b) self-monitoring; (c) home-school communication; and (d) positive reinforcement.

Working Together: Building Children's Social Skills Through Folk Literature. Corkum et al. (2010) explored the effectiveness of a SST manualized program, Working Together: Building Children's Social Skills through Folk Literature. The intervention was a program conducted in group sessions that involved the presentation of a folktale and its relationship to a specific social skill. The procedures involved discussion, practice, activities, feedback, contingency management, and role-play.

MOSAIC. Mikami et al. (2013, 2022) evaluated the effectiveness of the social skills program, Making Socially Accepting Inclusive Classrooms (MOSAIC). Mikami et al. (2022) compared MOSAIC to business as usual. MOSAIC is a program conducted in group sessions and involved two components: (a) teacher modeling of warm positive relationships with children and highlighting child strengths in front of peers to encourage peer acceptance, (b) classroom behavior management strategies (token economy, points for social skill demonstration), and (c) brief daily social skills lessons. The procedures for the social skills training included a class discussion of a specific skill, role-play of the skills, and reinforcement throughout the day for demonstrating the skill.

Multimodal. Multimodal programs are defined as those designed to improve multiple skills, such as organization, study skills, homework completion, academic performance, and social skills.

Challenging Horizons Program (CHP). Seven studies investigate effects of the Challenging Horizons Program (Evans et al., 2004, 2005, 2007, 2011, 2016, 2023; Schultz et al., 2017), designed for middle and high school students

Table 1. Characteristics and Demographics of the Included Studies.

		Design		P (*n*)		G (*n*)		School		Ethnicity (*n*)							
Authors	Inter	Type	Setting	T	C	M	F	Type	Grade	W	B	A	NA	O	M	L	Non-L
Barney et al. (2022)	STP-A	BG	A	218	107	241	84	Middle/High	6&9	NR	57	NR	NR	32	NR	237	88
Colton and Sheridan (1998)	SST-CBC	SC-MP	School	3	NA	3	0	Elem	NR	3	0	0	0	0	0	0	NA
Corkum et al. (2010)	SST-Folk Literature	WG	School	16	NA	10	6	Elem/Middle	3-6	16	0	0	0	0	0	0	0
Evans et al. (2004)	CHP	WG	A	7	NA	5	2	Middle	6-8	NR	NR	NR	NR	NR	NR	NR	NR
Evans et al. (2005)	CHP	WG	A	27	NA	21	6	Middle	NR	27	0	0	0	0	0	0	NA
Evans et al. (2007)	CHP-C	BG	School	42	37	0	18	Middle	6-8	0	0	0	0	5	0	0	NA
Evans et al. (2011)	CHP	BG	A	31	18	0	0	Middle	6-8	0	0	0	0	0	0	0	NA
Evans et al. (2016)	CHP CHP-M	BG	A/School	222	104	232	94	Middle	6-8	252	39	0	0	8	27	10	316
Evans et al. (2023)	CHP	BG	School	92	94	147	39	High	9-11	138	27	2	0	9	0	19	-19
Mikami et al, (2013)	SST-MOSAIC	BG	A	24	113	66	71	Elem	1-3	0	0	0	0	0	0	0	NA
Mikami et al. (2022)	MOSAIC	BG	School	291	267	-275	275	Elem	K - 5	223	87	110	4	4	130	83	409
Pfiffner et al. (2011)	CLS	WG	School	37	NA	29	8	Elem	2-5	35	5	7	1	0	10	1	NA
Pfiffner et al. (2016)	CLS	BG	School	72	63	96	39	Elem	2-5	36	12	28	0	0	27	32	0
Pfiffner et al. (2023)	CLS-R	WG	School	36	NA	23	13	Elem	2-5	15	2	4	1	3	7	13	23
Schultz et al. (2017)	CHP	BG	A	112	104	156	60	Middle	6-8	166	0	23	0	6	21	4	0
Sibley et al. (2011)	STP-A	SC-MP	A	1	NA	1	0	High	10	166	0	0	0	0	0	0	NA
Villodas et al. (2013)	CLS	WG	School	57	NA	40	17	Elem	2-5	23*	7*	8*	1	12*	6*		NA

Note: Inter = intervention; P = participation; G = gender; Elem = elementary; SST-CBC = Social Training Program – Conjoint Behavioral Consultation; STP-A = Summer Treatment Program for Adolescents; CLS = Collaborative Life Skills Program; CHP = Challenging Horizons Program ; CHP-C = Challenging Horizons Program - Classroom; CHP-FCU = Challenging Horizons Program with Family Check Up; CHP-AS = Challenging Horizons Program – After School; SST = social skills training; MOSAIC = Making Socially Accepting Inclusive Classrooms; SC-MP = Student Check-Up Mentoring Program; W = White; B = Black; A = Asian; NA = Native American; O = other; M = multiple; L = Latino/a/x; Non-L = Non Latino/a/x.

Table 2. Outcomes from Single-Case Design Studies.

						Tau-U			BC-SMD		
Authors (Year)	Outcome	*n*	Type	Type of measure	VA	ES	95% CI	Qual[a]	ES	95% CI	Qual
Colton and Sheridan (1998)	Isolated Play	3	SA	Direct observation	M	.72[b]	[−1.00, 0.43]	M	−0.99	[−1.78, −0.20]	M*
Colton and Sheridan (1998)	Negative Social Interactions	3	SA	Direct observation	N	.56	[−0.88, −0.27]	S	−0.60	[−1.66, 0.45]	M*
Colton and Sheridan (1998)	Positive Social Interactions	3	SA	Direct observation	M[b]	.80	[0.47, 1.00]	M	1.32	[0.58, 2.05]	L
Sibley et al. (2011)	Inappropriate Behavior (intervals)	1	M	Direct observation	[a]	.86	[−1.00, 44]	M	N/A	N/A	N/A
Sibley et al. (2011)	Negative Behavior/Business Meetings (frequency)	1	M	Direct observation	[a]	.33	[−.75, 0.082]	S	N/A	N/A	N/A
Sibley et al. (2011)	Negative Behavior (Recreation)	1	M	Direct observation	[a]	.20	[−.17, 0.57]	S	N/A	N/A	N/A

Note. *n* = number of students; SA = Stand-Alone, M = Multimodal, N = negligible; S = small; M* = moderate; L = large; Qual = qualitative description of effect size.
[a]Insufficient experimental control to determine functional relation.
[b]Functional relation was established.

with ADHD. Various versions of the CHP were investigated including an after-school and a teacher consultant model. The CHP includes individual and group education groups, recreation time, interpersonal skills group/training (ISG), contingency management, and parent consultation/training. Social competencies were the focus of ISG and are generalized throughout the program and recreation activities. Specifically, ISG was developed to help youth with ADHD understand the connection between their behavior and the reactions of others and the impressions individuals have of them. Procedures for the intervention involved: (a) instruction in problem solving steps and vocabulary related to ISG; (b) the development of "IDEAL self-goals," or goals for how they want others to perceive them; and (c) games and activities during, in which students and teacher rate the student's behavior to their IDEAL self-goals.

Collaborative Life Skills Program (CLS). Four studies investigated the effects of the CLS program (Pfiffner et al., 2011, 2016, 2023; Villodas et al., 2013), a program designed for students with ADHD in elementary schools. CLS includes a teacher component, parent component, and child skills component with a focus on collaboration across settings. The teacher component consists of behavior modification consultation, a school-home daily report, a homework-plan, and classroom accommodations. The parent component consisted of parent training modules focused on elements, such as improving homework routines, organization, peer interactions, stress management, use of praise, commands, rewards, discipline, and home-based token economy. The child component consisted of social skills modules (i.e., being a good sport, accepting consequences, assertion, dealing with teasing, problem solving, self-control, friendship making) and independence (homework skills, chores/tasks, and routines). Procedures for skills training included: (a) didactic instruction, (b) behavior rehearsal, and (c) in vivo practice with feedback and reinforcement). Additionally, Pfiffner et al. (2023) described the development and testing of a remote adaptation of CLS (CLS-R). The CLS-R adapted the program by shifting the delivery of training and support for school mental health providers to a web-based format. Coaching and resources were provided remotely, and progress monitoring was conducted through an online system.

Summer Treatment Program (STP). Two studies investigated summer treatment programs (Barney et al., 2022; Sibley et al., 2011), which was originally designed by W. E. Pelham et al. (2005). The STP includes contingency management (token/point system, praise, effective commands, time out, daily report card) across academic and recreation settings, parent training, group problem solving, and SST (Fabiano et al., 2014). SST involves instruction in specific skills including communication, participation, validation, and cooperation and reinforcement of use of the skills throughout the day. Procedures for the group sessions include: (a) instruction, (b) modeling, (c) role-playing; and (d) practice. Additionally, during peer-focused activities, such as sports activities, in vivo teaching and reinforcement are used to increase generalization of the skills taught.

Similarities and Differences. When comparing overall components, the programs are very similar (see Figure 2). Across all programs, interventions included child/adolescent social skills training, in vivo practice, and reinforcement. Five programs included behavior management strategies, such as token economies, and teacher training or consultation. Four programs included parent training or consultation. The primary difference lies in the inclusion of skills other than social competence, including academic and organizational skills, in the three multimodal programs and not the stand-alone programs.

Additional differences can be found in implementation setting and methods of intervention design and implementation. CLS is implemented in schools during the academic day, CHP is implemented in an after-school program or as a consultation model during the academic day, and STP is implemented in a summer treatment program. All but CHP are designed for elementary school students. Differences are evident in the social skills approach. Specifically, social skills are taught through direct instruction in explicit modules in CLS, through self-awareness and social goal setting including IDEAL self-goals in CHP, and daily training in STP. Both CHP and STP include in-vivo instruction and reinforcement of social skills through recreation time periods. Differences can also be found in parent training, with structured weekly sessions in CLS, periodic sessions (e.g., monthly) in CHP, and intensive weekly sessions in STP. Additionally, differences can be found in behavior management strategies used. Both CLS and CHP rely on systems developed in schools when the program was implemented during the school day. CHP, when implemented in an after-school setting, and STP include intensive behavioral systems. Differences can be found in the interventionists. School mental health staff implement CLS; teachers, counselors, and researchers implement CHP; and training clinicians and paraprofessionals implement STP. Finally, the intensity of the programs varies from approximately 12 weeks (CLS) to semester of yearlong (CHP) to daily for 6 to 8 weeks (STP).

Outcomes

Overall, the studies were single-case design ($k=2$), one group, pre-post test design ($k=6$), and between subjects ($k=9$). There were 108 effect sizes, with 42 indicating no effect, 34 small effects, 20 moderate effects, and 12 large effects on social outcomes.

Component	Social Skills Training Program (Colton & Sheridan)	Working Together (Corkum et al.)	MOSAIC (Mikami et al.)	CLS (Pfiffner et al.)	CHP (Evans et al.)	STP (Pelham et al.)
Child Training / Social Skills Lessons	✓	✓	✓	✓	✓	✓
Parent Training / Consultation	✓			✓	✓	✓
Teacher Training / Consultation	✓		✓	✓	✓	✓
Behavioral Management (e.g., token economy)	✓		✓	✓	✓	✓
Academic / Organizational Skills				✓	✓	✓
In Vivo Practice / Reinforcement	✓	✓	✓	✓	✓	✓
Multimodal (more than social skills)			✓	✓	✓	✓
Elementary	✓	✓	✓	✓		
Middle/High					✓	
Elementary/Middle						✓
Effect Size	0.56 – 0.80	0.37 -0.69	0.04 – 0.26	0.40 - 0.49	0.07 – 0.70	0.17 – 0.86

Figure 2. Component comparison across programs and effect sizes.

Effects at Study Level

Single-Case Design Studies. Single-case design studies ($k=2$; Colton & Sheridan, 1998; Sibley et al., 2011) included four participants (ages 8–9 years old) and reported six outcomes all measured through direct observation (see Table 3). One study was implemented in analogous classroom setting (Sibley et al., 2011; STP) and one in an elementary school (Colton & Sheridan, 1998). We analyzed the data through visual analysis (see Ledford et al., 2018; Vannest et al., 2016), Tau-U (Parker et al., 2011), and between-case standardized mean difference (BC-SMD; Pustejovsky et al., 2014) for one study that met the requirements (Colton & Sheridan, 1998).

Results indicated that social skills training produced small to large effects. Using visual analysis, only one study (Colton & Sheridan, 1998) used a design with sufficient experimental control. Colton and Sheridan (1998) established a functional relation between a conjoint behavioral consultation treatment package and positive social interactions and playing in isolation with moderate effects. No functional relation was established for negative social interactions as baseline data indicated very minimal negative interaction.

One Group Pretest/Posttest Design Studies. Within-group studies ($k=5$; Corkum et al., 2010; Evans et al., 2004, 2005; Pfiffner et al., 2011; Villodas et al., 2013) included 117 participants and measured outcomes through five tools (CIS, IRS, SRSS/SSIS, ACES, and report cards). We calculated Cohen's d (see Table 4) and qualified effects as no effect (<0.20), small (0.21–0.49), moderate (0.50–0.79), or large (≥ 0.80; Cohen, 1988). We caution the reader to interpret the qualifications cautiously as Cohen (1988) suggested that effect sizes be interpreted within context.

Results indicated that across all measures (SRSS/SSIS, CIS, IRS, and report card), parents and teachers rated small ($k=9$) to moderate ($k=9$) effects of social skills training on broadband and narrowband ratings of social skills with one study finding no effect on one rating (i.e., math teacher ratings of social functioning with peers, Evans et al., 2005).

Meta-Analysis Outcomes. We included nine studies in the meta-analysis, which included 81 effect sizes and 1,340 total participants. The first model built was a random effects meta-analytic model that used the inverse variance weighted mean at the study level. The base model provided an estimated effect of $g=0.0585$ ($SE=0.09$; 95% CI [−0.14, 0.26]). The model indicated large heterogeneity in effect size estimates. The I^2 index was 60.18% and the Q-statistic was significant ($df=8$; $Q=18.385$; $p=.0185$). The standard deviation of effect sizes at the study level was large ($\tau=.19$) and the variance of effect size estimates (τ^2) was.0365. The

Table 3. Outcomes from One-Group Pretest/Posttest Design Studies.

Outcome	*n*	Type	Measure	Rater	Cohen's *d*	ES qual	Direction of outcome
Corkum et al. (2010)							
Social skills	16	SA	SRSS	Teacher	0.61	M	Improvement
Social skills	16	SA	SRSS	Parent	0.69	M	Improvement
Social skills	16	SA	SRSS	Student	0.37	S	Improvement
Evans et al. (2004)							
Playmates	77	M	CIS	Parent	0.54	M	Improvement
Siblings	7	M	CIS	Parent	0.54	M	Improvement
Parents	7	M	CIS	Parent	0.33	S	Improvement
Peers	7	M	CIS	Teacher	0.60	S	Improvement
Teacher	7	M	CIS	Teacher	0.20	S	Improvement
Evans et al. (2005)							
Relation with peers	7	M	IRS	Parent	0.70	M	Improvement
Relation with peers	7	M	IRS	Science Teachers	-0.66	M	Decline
Relation with peers	7	M	IRS	Math Teachers	-0.18	N	No effect
Pfiffner et al. (2011)							
Social functioning	37	M	SSIS	Parent	0.49	S	Improvement
Social functioning	37	M	SSIS	Teacher	0.43	S	Improvement
Pfiffner et al. (2023)							
Prosocial behavior	29	M	SDQ	Parent	-0.22	S	Improvement
Prosocial behavior	31	M	SDQ	Teacher	-0.41	S	Improvement
Peer problems	29	M	SDQ	Parent	0.52	M	Improvement
Peer problems	31	M	SDQ	Teacher	-0.13	N	No effect
*Villodas et al. (*2023*)*							
Social skills	57	M	SSIS	Parent	0.66	M	Improvement
Peer impairment	57	M	IRS	Parent	0.26	S	Improvement
Social skills	57	M	SSIS	Teacher	0.60	M	Improvement
Interpersonal skills	57	M	ACES	Teacher	0.48	S	Improvement
Peer impairment	57	M	IRS	Teacher	0.47	S	Improvement
Social development grades	57	M	Report Card	Teacher	0.82	M	Improvement

Note. SA = Stand-Alone, Mult = Multimodal, N = negligible; S = small; M = moderate; L = large; SRSS = Social Skills Rating System (Gresham & Elliott, 1990); CIS = Columbia Impairment Scale (Bird et al., 1993); IRS = Impairment Rating Scale (Fabiano et al., 2006); SSIS = Social Skills Improvement System Rating Scales (Gresham & Elliott, 2008); SDQ = Strengths and Difficulties Questionnaire (Goodman, 2001); ACES = Academic Competence Evaluation Scales (DiPerna & Elliott, 2000).

prediction interval (PI) also highlighted the heterogeneity in intervention effects, 95% PI [-0.32, 0.43].

The interpretative model was a multi-level, multivariate model with robust variance estimation providing a more precise estimate of effects. The interpretative model provided an estimated effect of $g = 0.0901$ ($SE = 0.053$; 95% CI [-0.015, 0.195]). This model also indicated heterogeneity was present. The Q-statistic was statistically significant ($df = 80$; $Q = 108.676$; $p = .018$); the standard deviation of effect sizes at the study level was still large $\tau^2 = .011$ ($\tau = .1037$), whereas within study the standard deviation was smaller $\tau^2 = .045$ ($\tau = .002$). The 95% PI also highlighted the heterogeneity in intervention effects, 95% PI [−0.13, 0.31].

The forest plot (see Figure 3) visually shows the variability in effect size estimates at the study level and effect size level (see Supplemental Figure 3). Furthermore, the empirical Bayes distribution of potential effect sizes show a bimodal distribution of effects (see Supplemental Figure 4).

To attempt to explore the nature of this variability, we explored five specific moderators. For each moderator test, we removed the intercept from the model to allow for easier interpretation.

No statistically significant differences were found based on whether the comparison condition was active or business-as-usual, $F(df1 = 2, df2 = 79) = 1.52$, $p = .225$. The mean estimate for studies with an active comparison condition was 0.12 (95% CI [−0.025, 0.26]) and for business-as-usual was 0.08 (95% CI [−0.035, 0.20]).

No statistically significant differences were found based on the individual providing the rating for the outcome measure, $F(df1 = 6, df2 = 76) = 2.21$, $p = .06$. The mean estimate

Table 4. CEC Quality Indicators.

Author(s)	CEC 1.0 Context and setting	CEC 2.0 Participants	CEC 3.0 Description of interventionist	CEC 4.0 Description of intervention	CEC 5.0 Measured and reported fidelity	CEC 6.0 Internal validity	CEC 7.0 Outcome measures/ dependent variables	8.0 Data analysis	Total	Meets CEC standards
Barney et al. (2022)	Yes	Yes	No	Yes	No	No	Yes	Yes	5	No
Colton and Sheridan (1998)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Corkum et al. (2010)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Evans et al. (2004)	Yes	Yes	Yes	Yes	No	Yes	No	Yes	6	No
Evans et al (2005)	Yes	Yes	No	Yes	No	No	Yes	Yes	5	No
Evans et al (2007)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Evans et al. (2011)	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	7	No
Evans et al. (2016)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	No
Evans et al. (2023)	Yes	yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Mikami et al. (2013)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Mikami et al. (2022)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Pfiffner et al. (2011)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Pfiffner et al. (2016)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Pfiffner et al. (2023)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	8	Yes
Schultz et al. (2017)	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	7	No
Sibley et al. (2011)	Yes	Yes	Yes	Yes	yes	No	Yes	Yes	7	No
Villodas et al. (2013)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	7	Yes

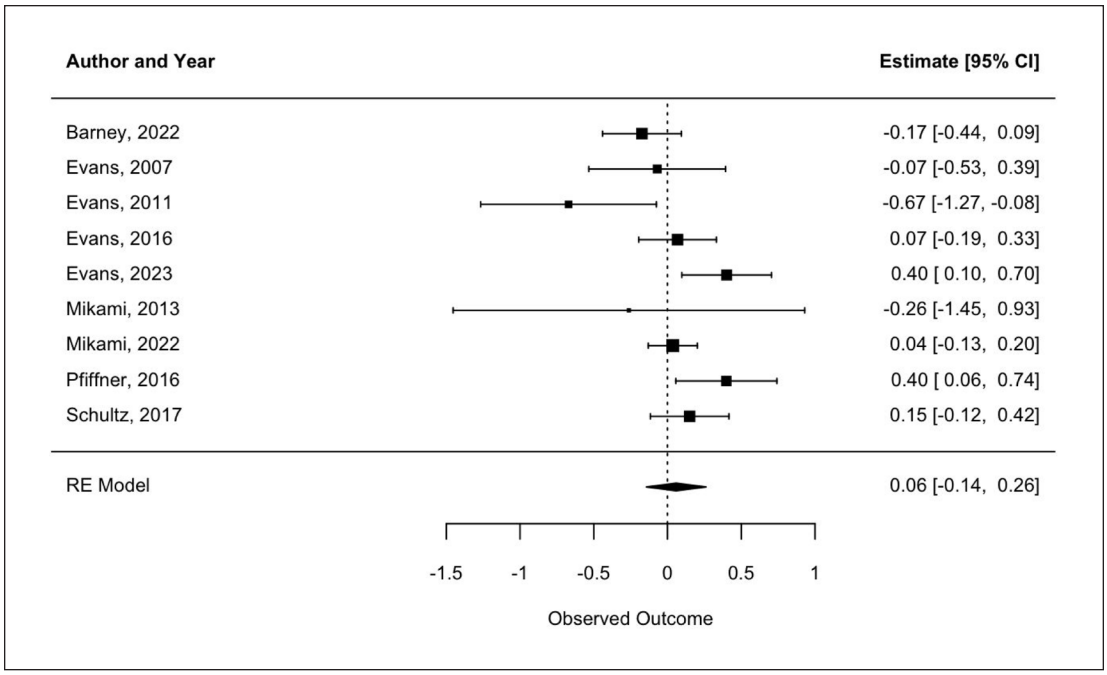


Figure 3. Forest plot of aggregated effect size at study level ($k = 9$).

for observation was −0.36 (95% CI [−0.955, 0.23]), for parent was 0.09 (95% CI [−0.03, 0.215]), for teacher was 0.12 (95% CI [-0.04, 0.27]), and for peer was −0.03 (95% CI [−0.255, 0.20]), and for student themselves was 0.265 (95% CI [0.026, 0.50]).

No statistically significant differences were found based on whether the outcome measure was considered broad or narrow, $F(df1 = 2, df2 = 79) = 1.32$, $p = .27$. The mean estimate for studies using broad measures was 0.08 (95% CI [−0.06, 0.22]) and for narrow band was 0.115 (95% CI [−0.009, 0.32]).

No statistically significant differences were found based on interventions categorized as stand-alone versus multimodal, $F(df1 = 2, df2 = 79) = 1.28$, $p = .28$. The mean estimate for interventions categorized as stand-alone was 0.10 (95% CI [−0.17, 0.38]) and for multimodal was 0.09 (95% CI [−0.04, 0.22]).

No statistically significant differences were found based on the age of participants, $F(df1 = 2, df2 = 79) = 1.53$, $p = .22$. The mean estimate for participants in elementary grades was 0.15 (95% CI [−0.065, 0.37]) and for secondary grades was 0.07 (95% CI [−0.06, 0.20]).

Publication Bias. We evaluated publication bias in two steps. First, we investigated two funnel plots; the first included all 81 effect size estimates (see Supplemental Figure 1), and the second included the weighted average of the nine studies (see Supplemental Figure 2). Our visual inspections of the plots led us to believe there is potential asymmetry in the effect sizes estimates. Second, we conducted an Egger's regression test that aligned with our visual analysis in suggesting asymmetry in the funnel plots ($z = 2.46$, $p = .01$; $b_0 = -0.003$; 95% CI [−0.013, 0.07]).

Discussion

This is the first systematic literature review and meta-analysis to explore the effects of school-based social skills interventions on the social outcomes of youth with ADHD. We included single-case design studies, within-group, and between-group studies. Although the meta-analytic effect for the nine between-group effects was negligible ($g = 0.09$), we found a large range of effect sizes from no effect to large effects. Our findings related to school-based social skills programs contribute to the knowledge base that previously included clinic-based (Powell et al., 2022; $d = 0.32$) and clinic and school-based group design studies (Morris et al., 2021; $d = 0.57$).

Across studies and 108 measured outcomes, effects were heterogenous, with 12 large effects, 20 moderate effects, 34 small effects, and 42 negligible or no effect. This suggests that it is possible that certain interventions and/or conditions may result in substantial improvements in specific contexts or with particular measures (Morris et al., 2021; Powell et al., 2022).

Due to the likely influence of the small number of studies, none of the moderating variables we tested statistically significantly influenced the model. Nonetheless, to offer

some perspective, we discuss several of the moderators we examined. First, the identity of the individual who rated child behavior seemed to differentiate outcomes with self-ratings indicating the largest effects, followed by teacher ratings, and then parent ratings. This pattern contradicts findings from Morris et al. (2021), who reported that parents perceived greater improvements than other raters. In addition to the small number of studies, the explanation for this discrepancy may be the context in which the interventions were delivered. For instance, Pfiffner et al. (2016) suggested that parent-reported improvements may be influenced when parents are aware of the intervention, a situation less likely to occur in schools than when interventions are delivered in clinical settings. Furthermore, because all interventions in this review were school based, it is likely that teachers had greater knowledge of the intervention content and implementation.

Nonetheless, this pattern did not hold across all studies. For example, Pfiffner et al. (2023) found that parent ratings reflected greater effects than teacher ratings. The authors proposed that teachers may be less able to observe social interactions in unstructured settings such as lunch and recess, limiting their capacity to detect change. The implications of these rater discrepancies remain unclear. It is possible that the difference could be contributed to the rating scales themselves. Evans et al. (2023) emphasized the importance of developing and using social outcome measures that are sensitive enough to detect intervention effects. Furthermore, regarding self-ratings, only one study contributed data (five effect sizes), so this finding should be interpreted with caution.

Additionally, in the current review outcomes were greater in elementary schools compared to middle or high schools. This finding contradicts previous research. For example, Evans et al. (2023) suggested that high school students with ADHD were more likely than middle school students to respond positively to social skills taught in CHP. They proposed that social skills interventions may be better suited for older adolescents. Given that social functioning often improves with maturity, it would be reasonable to expect stronger effects among high school students. This expectation may be particularly relevant when interventions require mature cognitive functioning, such as the IDEAL self-goals emphasized in CHP. However, this was not the case in the current review. It is possible that the age-related effects observed may be influenced by specific components of the intervention and/or characteristics of students, warranting further investigation.

Although the limited number of studies and insufficient information precluded an analysis of gender as a moderator, Mikami et al. (2013) disaggregated findings by participant gender and program (i.e., MOSAIC, COMET). Their results revealed that only boys (not girls) benefited from the Mosaic program relative to COMET. As the study relied on peer ratings, the authors hypothesized that girls with ADHD may be perceived as more deviant than boys with ADHD in general, as students are likely to be more exposed to boys with ADHD due to prevalence. As such, ratings of girls would be lower than those of boys. Consequently, peer perceptions of girls may have been more resistant to change, even when teachers modeled acceptance.

Additional research should explore the moderating effect of differences in programs. Although many similarities across programs were found, differences in social skill approaches, intensity, parent involvement, and settings were found. For example, interventions that provide opportunities to practice strategies learned in natural settings were more likely to show promising outcomes than those that did not. For example, Barney et al. (2022) found that the 360 hr of intervention, combined with scaffolded social activities, provided adolescents with ample opportunities to practice social skills and build reciprocal friendships. This type of in situ instruction, as was included in studies evaluated in this meta-analysis, where skills are practiced in authentic contexts, might be more effective than traditional, isolated instruction. Adolescents with ADHD often possess social knowledge, but struggle to apply it in everyday situations, making context-based practice essential for long-term improvement (de Boo & Prins, 2007). These findings emphasize the importance of providing opportunities for authentic practice in interventions, highlighting the broader factors that can influence the success of school-based social skills programs for students with ADHD. Unfortunately, the sample in this review was too small to make those comparisons. As more studies are conducted, it will be possible to explore specific differences in programs.

Additionally, with time, as research continues to explore the interventions discussed in the current study, and others that might be developed, it is important that this review be replicated exploring the impact of duration and intensity on student outcomes. Future studies should examine how length of the intervention and time required influence the effectiveness of social skills interventions for youth with ADHD. Understanding these relationships will help guide practitioners in selecting and implementing interventions that are both developmentally appropriate and contextually feasible.

Finally, differences in the effectiveness of social skills training may be attributed to individual child characteristics. Children with ADHD represent a heterogeneous population, and specific characteristics likely impact their response to intervention. For example, the severity of core ADHD symptoms may influence a child's ability to acquire and generalize social skills. Comorbid conditions, such as oppositional defiant disorder or anxiety may further complicate social functioning and lessen intervention effects. Additionally, cognitive abilities, including verbal skills, working memory, and emotion regulation, likely facilitate

the acquisition and application of new social behaviors. A child's awareness of their own social challenges may also play a role, with greater insight potentially leading to increases in engagement and receptivity to intervention. Finally, intrinsic motivation to improve peer relationships may enhance the impact of social skills training. Together, these factors underscore the importance of considering individualized approaches to social skills interventions.

An important finding in this review is the limited availability of school-based interventions specifically targeting social skills challenges among students with ADHD. Although only a small number of such interventions met inclusion criteria, it is encouraging that several have been evaluated across multiple studies and adapted in various formats. For example, multimodal interventions like the Summer Treatment Program (STP; Barney et al., 2022; W. E. Pelham et al., 2005; Sibley et al., 2011), Challenging Horizons Program (CHP; Evans et al., 2004, 2005, 2007, 2011, 2016, 2023; Schultz et al., 2016), and Collaborative Life Skills (CLS; Pfiffner et al., 2011, 2016, 2023; Villodas et al., 2013) have been tested across diverse settings and delivery models. The repeated evaluation of these interventions highlights their potential for scalability, adaptability, and sustained implementation in real-world school environments. However, it is also evident that the field remains largely dominated by these three multimodal programs, all of which have scholarly or conceptual ties to the original Summer Treatment Program (W. E. Pelham et al., 2005). However, we encourage researchers to consider findings across studies and potential moderators to continue expanding on these interventions and considering alternative approaches to supporting the social development of students with ADHD.

Although only a small number of school-based interventions for social skills challenges of students with ADHD were included, it was encouraging that several have been evaluated in multiple studies with varying versions. For instance, multimodal interventions, such as the Summer Treatment Program (STP), Challenging Horizons Program (CHP), and Collaborative Life Skills (CLS) have been tested across different settings and formats. The repeated evaluation of these interventions underscores their potential for scalability and adaptation. The field appears to be dominated by these three multimodal interventions (STP; CHP; and CLS) that share similar overlapping theoretical orientations (i.e., cognitive behavioral) and practices. Nonetheless, they include different content and were developed separate from each other, limiting what can be concluded when comparing the results of each.

Future research, following a design-based research model could evaluate the effects of including specific components based on ongoing assessment of student needs and responsiveness to intervention (e.g., Anderson & Shattuck, 2012). Consider the student with ADHD who demonstrates strengths in verbal reasoning but struggles with emotional regulation; the intervention could be tailored to capitalize on their verbal abilities through discussion-based strategies while simultaneously incorporating explicit instruction and practice in emotional regulation skills. Similarly, a student with strong motivation to form peer relationships but pronounced impulsivity may benefit from interventions emphasizing self-monitoring and reinforcement for behavioral inhibition. By embedding flexibility and responsiveness into intervention frameworks, researchers and practitioners may better address the heterogeneity of ADHD presentations and improve the effectiveness of social skills training efforts.

Limitations

Several limitations should be noted in this meta-analysis that may have affected the interpretation and generalizability of the findings. First, the limited number of studies specifically focused on school-based social skills interventions reduces the robustness of the conclusions drawn. Although we identified 17 studies, only 9 group design studies met the criteria for the meta-analysis. This limited sample size might diminish the statistical power or increase the likelihood that the effect sizes may not fully capture the variability of school-based social skills interventions across different educational settings and student populations. Another limitation is the variability in outcome measures across the included studies. Although we examined the effects of both broad and narrow measures of social functioning, the inconsistency in measurement tools and the reliance on self, peer, parent, and teacher ratings present challenges.

Another limitation was that the studies reviewed varied significantly in the intensity, duration, and fidelity of intervention implementation, which may have contributed to the heterogeneity in the results. Some interventions were implemented over several weeks or months, while others spanned an entire academic year or summer program, making it difficult to assess the impact of the intervention application on outcomes.

Additionally, not all studies reported fidelity checks or adherence to the intervention protocols, further complicating the ability to assess the actual effectiveness of the interventions. Future research should prioritize detailed reporting on implementation fidelity and focus on developing measurement tools to improve the consistency and comparability of outcomes within school settings.

Finally, we did not include the prior terms used to describe ADHD, such as ADD, Hyperkinetic Reaction of Childhood, or AD/HD in our Boolean string. As such, some earlier studies might not have been included. Nonetheless, we conducted an exploratory brief search and did not find any additional studies.

Conclusion

This systematic literature review and meta-analysis offers a comprehensive evaluation of the available data on the effectiveness of school-based social skills interventions (SST) for students with ADHD, addressing a crucial gap in the literature. Although the overall effect of these interventions are modest, specific programs, particularly multimodal ones, show significant promise by incorporating practice in educational settings (Barney et al., 2022; Evans et al., 2004; Pfiffner et al., 2016). Established interventions like the Summer Treatment Program (STP), Challenging Horizons Program (CHP), and Collaborative Life Skills (CLS) have been widely studied, providing valuable insights into how these programs can be effectively implemented in school settings to address the social skills deficits faced by students with ADHD (Evans et al., 2016; Pfiffner et al., 2011; Schultz et al., 2016). However, this study also highlights important challenges. Additionally, variability in outcome measures across studies complicates direct comparisons of effectiveness.

Moving forward, improving the rigor of methodologies through detailed reporting on implementation fidelity and the development of standardized measurement tools will enhance the consistency and comparability of results. Addressing these gaps will provide educators, administrators, and policymakers with more precise guidance on designing, implementing, and evaluating social skills interventions that promote long-term social and academic success for students with ADHD, ultimately contributing to more supportive and inclusive school environments.

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Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supplemental Material

Supplemental material for this article is available online.

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*Denotes studies included in the systematic review

**Denotes studies included in the meta-analysis

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